

The Top Open Source Hypervisor Technologies

A hypervisor monitors the creation and running of virtual machines. It allows a large number of machines with different operating systems to run on a single host machine. This article covers the types of hypervisors with descriptions of the top hypervisor technologies.



A hypervisor is a software layer which provides the capability to run multiple virtual machines on the same physical host. It generally falls into either one of the following two classifications — Type1 and Type2 hypervisors.

Type1 hypervisors directly run on the physical hardware. They control the hardware as well as manage the virtual machines. Type1 are also termed bare metal hypervisors.

Type2 hypervisors run as an application on an existing operating system (also known as the host operating system), which is installed on the bare metal. Here, there is an added complexity of the guest operating system calls needing to traverse via the host operating system stack before they reach the hardware.

Let's now look at a list of popular free hypervisors, both Type1 and Type2, which can be used by systems administrators. Hopefully, this will help them in choosing the most relevant hypervisor technology based on their requirements and will also help in evaluating other commercial hypervisors.

Type1 hypervisors

Linux KVM

A KVM (kernel based virtual machine) is a GNU/Linux based project developed for x86 machines. It operates with a loadable kernel module named *kvm.ko*, which turns the